

• With the development of the general theory of sets, the third stage began (the 20th century) when the concept of a function formerly defined only for numerical sets was generalized over the sets of an arbitrary nature.

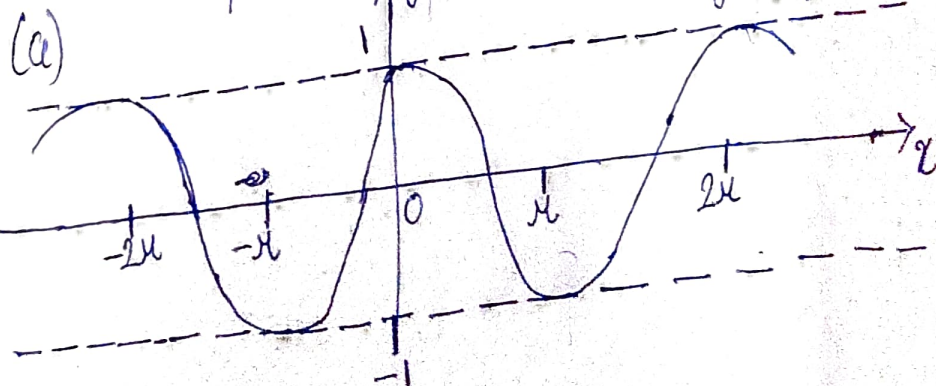
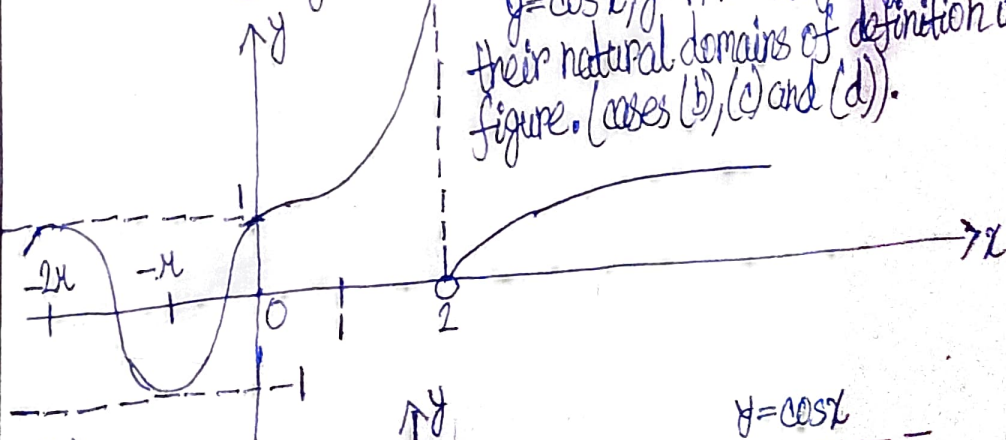
Reader: It appears that by overestimating the role of a formula we inevitably slip back to the concepts of the 18th century.

San: For further reading on the history of functions, see the sections on References and Further reading on the Wiki page "History of the function concept".

Author: Let us discuss now one more way of defining a function, namely, the graphical method. The graph of a function $y=f(x)$ is a set of points on the plane (x, y) whose abscissas are equal to the values of the independent variable (x) , and whose ordinates are the corresponding values of the dependent variable (y) . The idea of the graphical method of defining a function is easily visualized. Figure 14a plots the graph of the function

$$y = \begin{cases} \cos x, & x < 0 \\ 1+x^2, & 0 \leq x \leq 2 \\ \log(x-1), & x > 2 \end{cases} \quad \text{discussed earlier.}$$

For a comparison, the graphs of the functions $y = \cos x$, $y = 1+x^2$ and $y = \log(x-1)$ are shown within their natural domains of definition in the same figure. (cases (b), (c) and (d)).



(b)